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Genotype-by-environment interactions due to antibiotic resistance and adaptation in *Escherichia coli*

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Abstract

Mutations that are beneficial in one environment can have different fitness effects in other environments. In the context of antibiotic resistance, the resulting genotype-by-environment interactions potentially make selection on resistance unpredictable in heterogeneous environments. Furthermore, resistant bacteria frequently fix additional mutations during evolution in the absence of antibiotics. How do these two types of mutations interact to determine the bacterial phenotype across different environments? To address this, I used *Escherichia coli* as a model system, measuring the effects of nine different rifampicin resistance mutations on bacterial growth in 31 antibiotic-free environments. I did this both before and after approximately 200 generations of experimental evolution in antibiotic-free conditions (LB medium), and did the same for the antibiotic-sensitive wild type after adaptation to the same environment. The following results were observed: (i) bacteria with and without costly resistance mutations adapted to experimental conditions and reached similar levels of competitive fitness; (ii) rifampicin resistance mutations and adaptation to LB both indirectly altered growth in other environments; and (iii) resistant-evolved genotypes were more phenotypically different from the ancestor and from each other than resistant-nonevolved and sensitive-evolved genotypes. This suggests genotype-by-environment interactions generated by antibiotic resistance mutations, observed previously in short-term experiments, are more pronounced after adaptation to other types of environmental variation, making it difficult to predict long-term selection on resistance mutations from fitness effects in a single environment.

Introduction

In the absence of antibiotics, the phenotypic effects of resistance mutations vary with environmental conditions, such as nutrient status or temperature (Ryu, 1978; Jin & Gross, 1989; Perkins & Nicholson, 2008). The resulting genotype-by-environment interactions (Bataillon *et al.*, 2011; Trindade *et al.*, 2012) are potentially important for understanding selection on resistance. However, the phenotypic effects of resistance mutations are frequently modified by subsequent mutations at other sites (e.g. Cohan *et al.*, 1994; Schrag *et al.*, 1997; Reynolds, 2000; Paulander *et al.*, 2007), such as those that fix during adaptation in the absence of antibiotics. Given that resistant bacteria can face rapidly changing selection pressures from heterogeneous host or natural environments, understanding how these different types of mutations interact to determine the phenotype across different conditions is important for predicting the long-term fate of resistant lineages.

Previous work with *Escherichia coli* shows that both antibiotic resistance mutations (Trindade *et al.*, 2012) and beneficial mutations that fix during adaptation to a novel resource environment (Travisano & Lenski,

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1995, 1996; Ostrowski *et al.*, 2005) can indirectly alter fitness in other environments, but the net effect of having both types of mutations is not clear. For example, if mutations that ameliorate the fitness costs of resistance mutations in the present environment also reduce their indirect effects on fitness in other environments, restoring the wild-type multivariate phenotype, then selection against costs of resistance will erode genotype-by-environment interactions. Alternatively, new beneficial mutations may have their own indirect effects on the phenotype in other environments that are independent of those of the original resistance mutations. In this scenario, predicting how the interaction between new beneficial mutations and resistance mutations will influence the phenotype across different environments is more difficult. This potentially prevents meaningful predictions for long-term selection on resistance mutations based on observed fitness effects in the presence of drugs or in a single drug-free environment.

I aimed to test for genotype-by-environment interactions associated with antibiotic resistance mutations both before and after adaptation to an antibiotic-free environment and to compare the effect of adaptation to that observed in the antibiotic-sensitive ancestral genotype. I isolated nine rifampicin-resistant genotypes of *E. coli* K12 MG1655, each with a different mutation in *rpoB,* which encodes the β-subunit of RNA polymerase, the target enzyme of rifampicin (Severinov *et al.*, 1993; Pozzi *et al.*, 1999; Campbell *et al.*, 2001; Trinh *et al.*, 2006). I then tested for costs of resistance and for fitness recovery during evolution in the absence of antibiotics. I did this by measuring competitive fitness relative to the ancestral strain in LB growth medium, both before and after approximately 200 generations of growth. Finally, I measured the ability of each genotype to grow in 31 different single-carbon-source environments (Perkins & Nicholson, 2008) and did the same for control genotypes that had no rifampicin resistance mutations but had adapted to the same experimental environment (LB). Thus, my experiment comprised four types of bacteria: wild type (no resistance mutation, no experimental evolution), resistant (defined *rpoB* mutation, no experimental evolution), resistant-evolved (defined *rpoB* mutation, approximately 200 generations experimental evolution) and sensitive-evolved (no resistance mutation, approximately 200 generations experimental evolution). I found that *rpoB* mutations and evolution in LB both altered the bacterial phenotype across many environments, but resistant-evolved genotypes were more phenotypically different from the wild type than either resistant or sensitive-evolved genotypes, demonstrating that adaptation in the absence of antibiotics does not ameliorate the indirect effects of antibiotic resistance mutations on growth in unselected environments.

Materials and methods

Bacterial strains and growth conditions

I used *E. coli* K12 MG1655 in all experiments, grown at 37 °C in liquid LB medium with or without rifampicin, or in single-carbon-source media in EcoPlates (Biolog, Hayward, CA,USA).

To isolate rifampicin-resistant mutants, I grew 480 independent replicate cultures of wild-type bacteria, each in 150 μL of antibiotic-free LB, for 24 h. I then spotted 5 μL of each culture onto agar plates containing 50 mg L^{-1} of rifampicin. After 24-h incubation, I picked 16 colonies at random from 16 different cultures, restreaked them to purify genotypes and confirm heritable resistance, and grew them in liquid LB for 3 h prior to storage at −80 °C in 25% v:v glycerol.

To identify rifampicin resistance mutations, I sequenced the central rifampicin resistance region of *rpoB* for each colony isolate. Primers for polymerase chain reaction (PCR) amplification were *fwd* 5′-CGTC GTATCCGTTCCGTTGG-3′ and *rev* 5′-TTCACCCGGAT AACATCTCGTC-3′ (Trindade *et al.*, 2009). I used the same primers for forward and reverse sequencing. I identified nine different amino acid changes in *rpoB*, all of which have been previously associated with rifampicin resistance in *E. coli* (Reynolds, 2000; Garibyan *et al.*, 2003; Trindade *et al.*, 2009).

Experimental evolution in LB

Selection lines were initiated with each of the nine different *rpoB* mutants and evolved for approximately 200 generations in the absence of antibiotics; I simultaneously maintained five replicate populations of the wild type in identical conditions. This was done in 100 μL liquid LB at 37 °C in 96-well microplates. Approximately every 12 h, I diluted each population 1000-fold into fresh media (approximately 1.4×10^5 cells), allowing for approximately 10 generations per growth cycle, and continued this for 20 transfers before isolating a single clone at random from one population initiated with each *rpoB* genotype and each of the five control populations; genotypes were then stored at −80 °C in 25% v:v glycerol. Thus, I isolated two genotypes for each of the nine *rpoB* mutations: one that was stored at −80 °C upon isolation (resistant) and one that had evolved for 200 generations in LB (resistant-evolved). I also isolated five control genotypes that had no resistance mutations but evolved in the same conditions (sensitive-evolved).

Rifampicin-resistant genotypes can fix secondary mutations on *rpoB* during adaptation in antibiotic-free conditions (Reynolds, 2000; Hall *et al.*, 2010), and mutations in RNA polymerase subunits, such as that encoded by *rpoB*, can be important in adaptation to novel growth media (Herring *et al.*, 2006; Applebee *et al.*, 2008; Conrad *et al.*, 2010; Tenaillon *et al.*, 2012).

I therefore resequenced the same region of *rpoB* for each of the evolved genotypes, but found no additional mutations for either resistant-evolved or sensitive-evolved genotypes. Note that the sequenced region does not cover the entire gene, and therefore, I do not exclude the possibility of mutations elsewhere on *rpoB* or on other RNAP subunits. I also confirmed that resistance to rifampicin was retained after evolution in LB by plating on LB agar supplemented with 50 mg L^{-1} rifampicin.

Competitive fitness assays in LB

To test for costs of resistance and for adaptation to LB, I measured fitness of the wild type and each resistant, resistant-evolved and sensitive-evolved genotype by competition assays in antibiotic-free LB against a marked strain of the wild type: *E. coli* K12 MG1655 Δara, which is otherwise isogenic and forms red colonies on tetrazolium arabinose (TA) agar (tryptone 1%, yeast extract 0.1%, NaCl 0.5%, L(+)arabinose 1%, TTC 0.005%), whereas all other genotypes form white colonies. For each competition, I grew independent cultures of both genotypes overnight in liquid LB at 37 °C, before mixing them 1 : 1 by volume, diluting 1000-fold into fresh LB and passaging for two growth cycles using the same protocol as during experimental evolution. I estimated the frequency of each competitor at the start and end by plating on TA agar. I calculated the selection coefficient s as the per generation difference in malthusian parameters between the two genotypes in each competition, where $s = \ln(R_{\mathrm{final}}/R_{\mathrm{initial}})/t$, where R_{final} and R_{initial} are the ratios of the competitors at 24 and 0 h, respectively, and t is assay duration in generations (Lenski *et al.*, 1991; Trindade *et al.*, 2009; Gullberg *et al.*, 2011). I ran three independent replicates for every genotype, plus eight replicates for the wild type, and discounted each score by the cost of the Δara marker (Trindade *et al.*, 2009), which was marginal (0.005).

Biolog assays

I measured the ability of every genotype to grow in each of 31 single-carbon-source environments on EcoPlates (Biolog). For each assay, I reconditioned bacteria from frozen by overnight growth in 150 μL LB. I then removed residual nutrients and starved the cells by centrifuging at 5000 *g* for 5 min, resuspending in phosphate-buffered saline (Cooper & Lenski, 2000), diluting 200-fold into fresh buffer and incubating for at least 2 h at 37 °C. To begin the assay, I added 150 μL of starved culture to each of 32 wells of a prewarmed EcoPlate, each containing a different carbon substrate or water. I measured OD_{600} immediately and after 24-h incubation at 37 °C, quantifying growth score as $\Delta OD_{24\ h}$, the change in optical density corrected by subtracting the score for the same genotype on water on the same microplate (MacLean & Bell, 2003; Hall *et al.*, 2011). I ran three replicate assays for every genotype, where each assay was initiated from an independent overnight culture, and 15 independent assays for the wild type, randomly assigning genotypes and replicates across microplates.

Escherichia coli can only use a subset of the substrates on EcoPlates. I therefore categorized a substrate as nonviable for a given genotype if the mean score ($\Delta OD_{24\ h}$) was less than 0.05; this is an arbitrary value that approximates the lower asymptote when growth of the wild type is plotted against substrates ranked by growth score. This excludes some growth scores that could be taken as significantly greater than zero by other statistical criteria, but I used a conservative cut-off for positive growth to avoid the increased measurement error associated with scores close to the detection limit of the spectrophotometer. I excluded eight substrates where no genotypes showed positive growth.

Statistical analyses

To test for adaptation of resistant genotypes to LB medium, I used a linear mixed-effects model with competitive fitness as the response variable, genotype (*rpoB* mutation) as a random effect and evolution (before or after approximately 200 experimental evolution) as a fixed effect. I then tested for variation of growth scores across single-carbon-source environments and among resistant genotypes by a model with growth score ($\Delta OD_{24\ h}$) as the response, genotype as a random effect and environment (carbon substrate) as a fixed effect. I also tested whether the phenotypic outcomes of evolution for resistant genotypes could be predicted by those for sensitive genotypes. I did this by measuring the correlation between changes in growth score following evolution ($\Delta\Delta OD_{24\ h}$) for resistant-evolved and sensitive-evolved genotypes across all environments.

To compare genotypes in terms of how their growth varies across different substrates (i.e. the relative qualities of different substrates), I calculated the pairwise Euclidean distance among all genotypes (MacLean & Bell, 2003). For a given pair of genotypes, this is the square root of the sum of squared differences across all environments. This incorporates differences in average score across all substrates (one genotype being better than the other on average) and differences in the relative qualities of different substrates (genotype-by-environment interactions). To discriminate between these two sources of phenotypic variation, I calculated a second distance matrix using the normalized growth scores for each genotype: after dividing the score for a given genotype on each substrate by the mean score for that genotype across all substrates, every genotype has a mean score of 1.0 and Euclidean distances can only be generated by differences between genotypes in the relative qualities of different substrates or by measurement error.

Finally, to visualize the position of each genotype in multivariate phenotypic space, I used principal components analysis, treating growth scores on different substrates as different variables and plotting the first two principal components derived from the covariance matrix, which incorporates differences in variance among variables. I used covariances because different variables had the same units, although I note that results were qualitatively unchanged using the correlation matrix. I used R and JMP for all analyses.

Results

Fitness effects of *rpoB* mutations before and after adaptation to LB medium

On average, *rpoB* mutations reduced bacterial fitness in the absence of antibiotics (mean $s \pm$ SD = -0.045 ± 0.039; Fig. 1). Following approximately 200 generations of growth in antibiotic-free medium, resistant-evolved genotypes had higher competitive fitness than the wild type (mean $s \pm$ SD = 0.087 ± 0.014; effect of evolution on fitness: $F_{1,8} = 81.23$, $P < 0.0001$). Antibiotic-sensitive genotypes that adapted to LB also had positive average fitness compared with the wild type (mean $s \pm$ SD = 0.091 ± 0.018; Fig. 1), with no significant variation among independently evolved replicates ($F_{4,9} = 2.08$, $P = 0.17$). Resistant-evolved and sensitive-evolved genotypes were no different in terms of average competitive fitness (Welch's *t*-test: $t_{6.8} = 0.43$, $P = 0.68$), resulting from a relatively large response to selection for resistant-evolved genotypes on average ($t_{11.6} = 2.55$, $P = 0.03$). In summary, both antibiotic-resistant and control lines adapted to LB and converged on similar fitness values at the end of the experiment.

Effects of resistance and adaptation vary among genotypes and environments

Rifampicin resistance mutations did not alter bacterial growth across single-carbon-source environments on average (mean $\pm$ SD difference from wild-type growth score = 0.01 ± 0.05, $P > 0.05$). However, different resistant genotypes varied considerably in their average growth score (effect of genotype: $F_{8,176} = 12.15$, $P < 0.0001$), with some having a higher average than the wild type, and others lower. Furthermore, the relative qualities of different environments varied among resistant genotypes (genotype $\times$ environment interaction: $F_{176,345} = 2.48$, $P < 0.0001$; Fig. 2a), indicating that bacteria with different *rpoB* mutations had distinct multivariate phenotypes. Note that this analysis excludes the wild type, but by definition, if there are

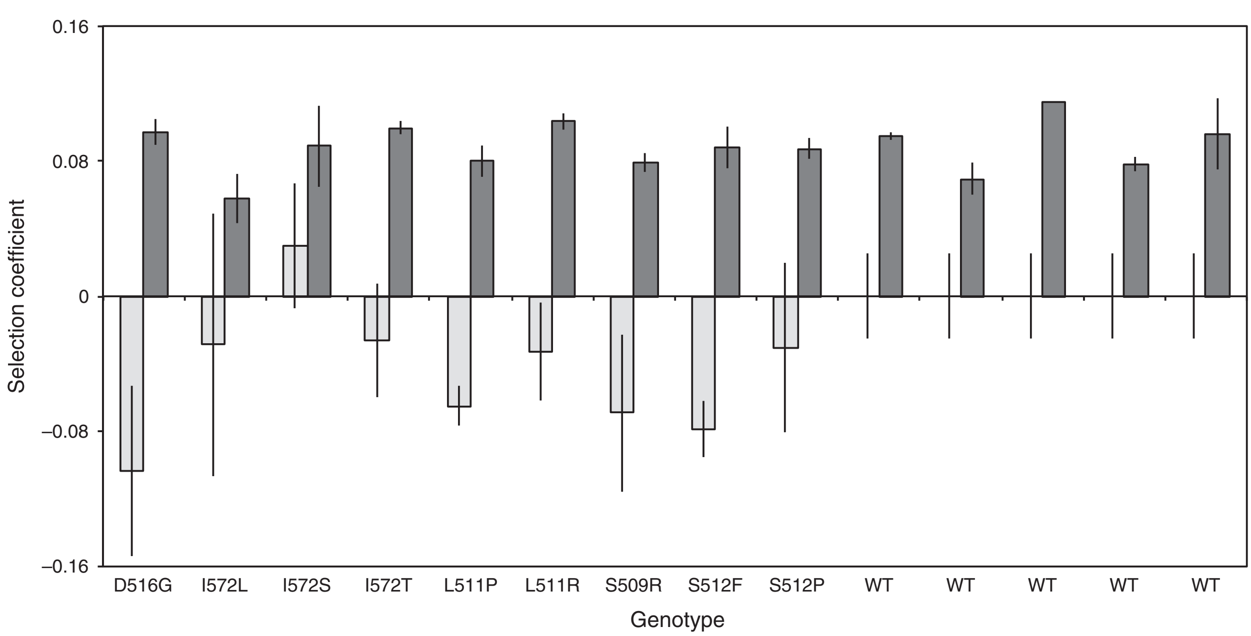

Fig. 1 Competitive fitness of rifampicin-resistant genotypes and wild-type *Escherichia coli* K12 MG1655 before (light bars) and after (dark bars) approximately 200 generations of evolution in LB growth medium. Each bar shows the average ($\pm$ SE) selection coefficient (s) for two or three replicates. Resistant genotypes are labelled by their amino acid changes on *rpoB*. The wild type (WT) before evolution has $s = 0$ and SE = 0.025 for eight replicates; five separate bars are shown for the wild type after evolution, representing independently evolved genotypes.

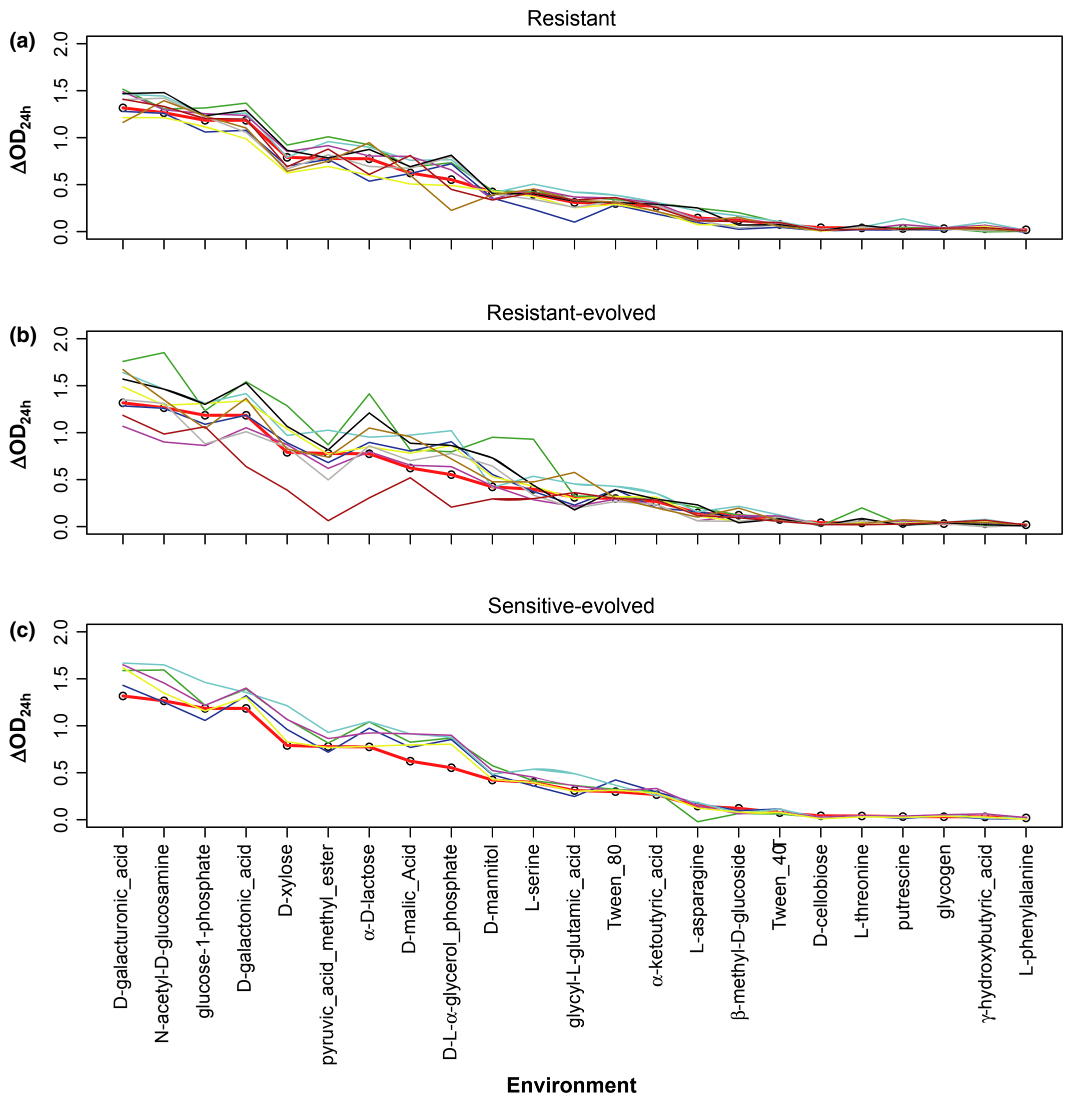

Fig. 2 Growth scores, measured as the change in OD_{600} over 24 h ($\Delta OD_{24\ h}$), across 23 different single-carbon-source environments for (a) resistant, (b) resistant-evolved and (c) sensitive-evolved genotypes. Substrates are ranked by growth of the wild type, which is shown by the thick red line in each panel (average of 15 replicates). Each other line shows the average score from three replicates for a different genotype.

genotype-by-environment interactions among resistant genotypes, then at least some of them also differ from the wild type. I analyse phenotypic differences from the wild type in more detail below.

Adaptation to LB did not alter the growth score of resistant genotypes averaged across all substrates ($F_{1,8} = 0.82$, $P = 0.39$), but the effect varied among both substrates (evolution × environment interaction: $F_{22,181} = 2.90$, $P < 0.0001$) and genotypes (evolution × genotype interaction: $F_{8,176} = 11.74$, $P < 0.0001$). Thus, the lack of an overall effect of evolution is caused by an increase in growth on some substrates for some genotypes and a decrease for others (Fig. 2b). By contrast, adaptation of antibiotic-sensitive bacteria to LB had a net positive effect on growth in single-carbon-source environments: sensitive-evolved genotypes had higher growth scores on average than the wild type (mean difference ± SD = 0.08 ± 0.04, $P < 0.05$; Fig. 2c).

I next tested whether the responses to selection across environments were correlated for resistant and sensitive bacteria (i.e. whether phenotypic evolution in resistant bacteria could be predicted by that in sensitive bacteria). Average changes in growth score after evolution ($\Delta\Delta OD_{24\ h}$) for resistant genotypes were positively correlated with those for sensitive genotypes in the

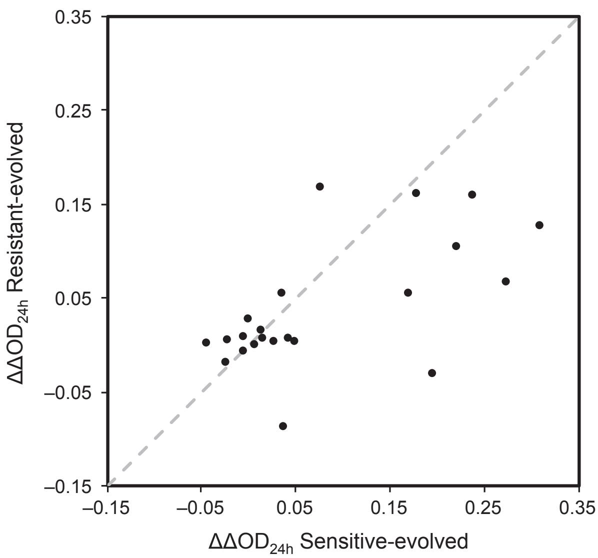

Fig. 3 Correlation of evolutionary changes in growth score in resistant-evolved and sensitive-evolved genotypes. Each point shows the average change in growth score ($\Delta\Delta OD_{24\ h}$) following adaptation to LB media for resistant-evolved and sensitive-evolved genotypes in a single environment (n = 23). The dashed line shows $y = x$.

Table 1 Phenotypic distances within and between different categories of bacteria. Distances are calculated as averages from all pairwise comparisons between genotypes within categories (n = 9 resistant, nine resistant-evolved and five sensitive-evolved), where the distance in each pairwise comparison is taken as Euclidean distance across all single-carbon-source environments. Standard error is given for each value.

	Resistant	Resistant-evolved	Sensitive-evolved
Resistant	0.57 ± 0.03		
Resistant-evolved	0.89 ± 0.04	1.05 ± 0.08	
Sensitive-evolved	0.71 ± 0.04	0.80 ± 0.06	0.49 ± 0.05
Wild type	0.42 ± 0.02	0.81 ± 0.11	0.68 ± 0.09

Table 2 Phenotypic distances, after accounting for differences in average growth score among genotypes. Distances are calculated as in Table 1 except that Euclidean distances are calculated from normalized growth scores, so that all genotypes have the same average score.

	Resistant	Resistant-evolved	Sensitive-evolved
Resistant	0.95 ± 0.05		
Resistant-evolved	1.37 ± 0.05	1.39 ± 0.13	
Sensitive-evolved	0.98 ± 0.04	1.04 ± 0.09	0.64 ± 0.04
Wild type	0.70 ± 0.07	1.17 ± 0.16	0.76 ± 0.04

same environments ($r^2 = 0.32$, $F_{1,21} = 9.75$, $P = 0.005$; Fig. 3). However, this relationship varied considerably among selection lines initiated with different resistant genotypes: $\Delta\Delta OD_{24\ h}$ for the wild type was a good predictor of $\Delta\Delta OD_{24\ h}$ for only three of nine resistant genotypes after accounting for multiple comparisons (Table S1). Thus, although evolution of the wild type loosely predicts the average change in growth score for resistant genotypes, it is a poor predictor for individual selection lines. Note that with a single selection line for each resistant genotype, variation of the effect of evolution among resistant-evolved genotypes is not necessarily due to an effect of starting genotype on evolution; stochastic processes during adaptation to LB could also contribute to variation among individual selection lines.

Phenotypic divergence due to resistance mutations and adaptation

Resistant-evolved genotypes showed a greater average phenotypic distance from ancestral bacteria than the same resistant genotypes prior to experimental evolution (paired t-test: $t_8 = 3.70$, $P = 0.006$, Table 1). Resistant-evolved genotypes were not significantly different from sensitive-evolved genotypes in terms of total Euclidean distance from the wild type (Welch's t-test: $t_{11.6} = 0.97$, $P = 0.35$).

Large phenotypic distances between genotypes can result from a difference in average score across all substrates or from differences in the relative qualities of different substrates. By normalizing the growth scores for each genotype on each substrate, I isolated the effect of the latter. These distances show that resistant-evolved genotypes are more phenotypically different from the wild type than either resistant ($t_8 = 2.79$, $P = 0.02$) or sensitive-evolved genotypes ($t_{9.1} = -2.48$, $P = 0.035$; Table 2). Furthermore, accounting for differences in average score reduces the difference between resistant and sensitive-evolved genotypes in terms of distance from the ancestor, because sensitive-evolved genotypes had higher average scores. Thus, resistance mutations and mutations that were beneficial in LB independently caused phenotypic divergence from the wild type to a similar extent in terms of changing the relative qualities of different carbon sources, but the largest effects were observed for bacteria with both types of mutations.

Next I compared the phenotypic distances within and between resistant, resistant-evolved and sensitive-evolved groups. The greatest distances were observed when comparing resistant-evolved genotypes to each other or to other genotypes (Tables 1 and 2). It is important to note that both types of phenotypic distance could also be generated by measurement error, in which case the relatively large distances for resistant-evolved genotypes would be associated with high between-replicate variance compared with the other groups. The intraclass correlation coefficient quantifies the repeatability of replicated measurements (Lessells &

Boag, 1987; Sokal & Rohlf, 1995), and in this experiment, it was not only high for each category, but was greatest for resistant-evolved bacteria (resistant: 0.94, resistant-evolved: 0.97, sensitive-evolved: 0.95, wild type: 0.82). I therefore exclude measurement error as an explanation for relatively high phenotypic divergence of resistant-evolved bacteria. In summary, resistant-evolved genotypes were the most distinct genotypes in comparison with wild type, resistant or sensitive-evolved genotypes; they were also the most diverse group in terms of having distinct growth profiles across all environments.

Finally, to visualize the phenotypic distances between genotypes across many environments, I used principal components analysis, which shows the same qualitative pattern as analysis of Euclidean distances above (Fig. S1): resistant-evolved genotypes lie farthest from the wild type in multidimensional phenotypic space and are also further away from each other than resistant or sensitive-evolved genotypes.

Discussion

Rifampicin resistance mutations incurred an initial fitness cost that was reduced during adaptation to antibiotic-free media. Consistent with previous studies, resistance mutations altered bacterial growth across single-carbon-source environments in terms of changing the relative qualities of different substrates for bacterial growth (Perkins & Nicholson, 2008; Bataillon *et al.*, 2011; Hall *et al.*, 2011) and so did adaptation to antibiotic-free media. The strongest effects were observed for bacteria with both types of mutations (resistance mutations and mutations associated with adaptation to LB), resulting in phenotypes that were more different from each other and from the wild type than resistant genotypes prior to adaptation and sensitive genotypes that adapted to the same experimental environment.

Why do *rpoB* mutations influence bacterial growth in single-carbon-source environments? Changing the structure of the β-subunit of RNA polymerase can have direct effects on transcription efficiency, and these contribute to costs of resistance (Reynolds, 2000). This may cause an average reduction in bacterial growth rate across all environments, but the consequences of altered transcriptional activity for fitness can also vary among environmental conditions (Hall *et al.*, 2011), meaning that direct effects on the function of the mutated enzyme can generate genotype-by-environment interactions even without having pleiotropic effects on other traits or the expression of other genes. Nevertheless, mutations in RNA polymerase do affect expression of many other genes (Applebee *et al.*, 2008; Conrad *et al.*, 2010; Derewacz *et al.*, 2013), and these indirect effects probably contribute to the changes in growth across antibiotic-free environments that were observed here and previously (Perkins & Nicholson, 2008; Hall *et al.*, 2011). Given that mutations in RNA polymerase can fix during adaptation to costs of resistance (Reynolds, 2000; Brandis *et al.*, 2012; Comas *et al.*, 2012) or to novel growth conditions (Applebee *et al.*, 2008; Conrad *et al.*, 2010), this may represent a common source of phenotypic divergence in populations evolving in heterogeneous environments.

It could be argued that the divergent phenotypes of resistant-evolved bacteria are simply a result of having a greater number of mutations than the other genotypes, and that adaptation to LB has a similar net effect on resistant and sensitive backgrounds. Indeed, because resistant genotypes were diverged from the wild type at the start of the experiment, the net change in normalized phenotypic distance from the ancestor following adaptation to LB was no different on average for resistant-evolved and sensitive-evolved genotypes ($t_9 = 1.65$, $P = 0.13$). However, adaptation to LB did appear to have different effects in resistant and sensitive bacteria, as shown by the poor correlation between changes in growth score across different substrates, suggesting that phenotypic changes in unselected environments do not simply accumulate additively with genetic distance from the ancestor.

Adaptation to LB caused an increase in bacterial growth on average (across all substrates) in sensitive-evolved genotypes. This is consistent with results from *E. coli* B, showing that adaptation to glucose-limited growth medium can increase fitness in other environments (Ostrowski *et al.*, 2005). By contrast, the change in bacterial growth averaged across all environments was marginal for resistant-evolved genotypes, despite considerable genotype-by-environment interactions. This suggests that the net effect of adaptation to a novel environment can be influenced by prior resistance evolution. The direct response to selection measured by the change in fitness in LB was relatively high compared with a previous experiment with the same strain in LB medium (Sousa *et al.*, 2012). A possible explanation is that my experiment was carried out in spatially static microcosms with a different transfer regime, potentially imposing selection for a different set of beneficial mutations.

Different outcomes of evolution for resistant and sensitive bacteria, in terms of the change in competitive fitness, average growth score across Biolog substrates and in their preferences for different carbon sources, could be caused by two mechanisms. First, different genotypes may acquire the same beneficial mutations, but their phenotypic effects are epistatic. Epistatic variation of fitness effects among resistant genotypes appears to be both common and strong (Rozen *et al.*, 2007; Trindade *et al.*, 2009), although it is unclear whether this also applies to pleiotropic effects. Second, resistant-evolved genotypes could fix different mutations to sensitive-evolved genotypes, because they occur at different frequencies or have different fitness effects, and indirect effects on growth in other environments are specific to different beneficial mutations. Consistent with this, several studies have shown that different resistant genotypes fix different beneficial mutations in the absence of antibiotics (Maisnier-Patin *et al.*, 2002, 2007; Hall *et al.*, 2010). Both of these mechanisms can be caused by epistasis, but disentangling the relative contributions of mutation frequency and epistasis is beyond the scope of this manuscript.

My results are consistent with evidence that the outcomes of evolution are contingent on starting genotype for rifampicin-resistant mutants (Barrick *et al.*, 2010; Hall *et al.*, 2010). However, I stress that with only one selection line for each resistant genotype, variation among resistant-evolved genotypes could also be due to stochastic variation in adaptive processes, such as the random appearance and loss of beneficial mutations (Travisano *et al.*, 1995) that could potentially cause individual selection lines to diverge even if they were initially identical. Therefore, the finding that resistant-evolved genotypes are phenotypically diverged from each other does not necessarily demonstrate an effect of starting genotype on adaptation. Furthermore, sensitive-evolved lines were all initiated with the wild type. Therefore, I cannot test whether the divergence among resistant-evolved genotypes would also apply among different antibiotic-sensitive genotypes that adapt to the same environment. An additional limitation is that I cannot exclude the possibility that resistant mutants acquired additional mutations during isolation on agar plates or reconditioning from freezer stocks by growth in liquid medium prior to the evolution experiment. Nevertheless, any strongly beneficial mutations during growth on rifampicin agar plates would have been visually detectable by heterogeneous colonies, and beneficial mutations during growth in liquid would likely reduce the phenotypic distance between resistant and resistant-evolved genotypes, making it unlikely that this explains the observed effects of adaptation.

Why does adaptation to LB influence bacterial growth in single-carbon-source environments? LB medium contains a diverse mixture of amino acids, which are the primary carbon sources in this media (Sezonov *et al.*, 2007). Therefore, a possible explanation is that selection for more efficient exploitation of amino acids in LB would increase bacterial growth on amino acids in single-carbon-source environments. This effect does not explain my results: there are four amino acids on EcoPlates, and the mean response to selection was not particularly high for any of them: L-serine, L-asparagine, L-phenylalanine and L-threonine were ranked 8th, 21st, 14th and 9th of 23 substrates, respectively, in terms of average $\Delta\Delta OD_{24\ h}$ across evolved genotypes. This type of analysis is probably too simplistic to reveal the effects of selection for improved growth in these conditions: *E. coli* has complex growth dynamics in LB, switching from sequential to simultaneous substrate use as nutrients are depleted and changing expression of genes involved in metabolism and growth in low-nutrient conditions (Baev *et al.*, 2006a,b). Therefore, adaptation may alter the expression of genes associated with metabolism of particular carbon sources as well as a general capacity for growth in low-nutrient environments. Finally, the finding that adaptation to novel growth media indirectly alters growth on other carbon sources is consistent with similar results from *E. coli* B, showing that adaptation to glucose-limited medium alters the phenotype across different types of minimal media (Cooper & Lenski, 2000; Cooper, 2002).

Environmental modulation of the phenotypic effects of antibiotic resistance mutations has been observed previously (Björkman *et al.*, 2000; Nagaev *et al.*, 2001; Paulander *et al.*, 2009; Trindade *et al.*, 2012). My results suggest that this is even more pronounced for resistant bacteria that undergo subsequent evolution in the absence of antibiotics, moving them into otherwise unexplored areas of phenotype space. This makes it difficult to predict the evolutionary fate of resistant bacteria based on fitness measurements in a single environment. For example, *in vitro* or *in vivo* fitness measurements may suggest that a given resistance allele will be under negative selection in populations of pathogenic bacteria, but fitness costs may vary with spatial and temporal heterogeneity of the host environment, and my results suggest that this variation can be amplified over evolutionary time.

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Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1 Correlation between responses to selection for selection lines initiated with rifampicin-resistant genotypes and the wild type.

Figure S1 Principal components analysis of bacterial phenotypes.

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